

Physics Frontier for AI-Driven Vehicle Dynamics Forecasting

The architectural ambition of the "PIT WALL" platform necessitates a profound leap in computational vehicle dynamics. The core problem—termed "kinetic hallucination"—arises because the foundation time-series model (IBM Granite TimeSeries TTM) forecasts telemetry channels independently, devoid of Newtonian coupling. To mitigate this, the forecasting pipeline employs a tri-layer architecture where the raw tensor is passed through a differentiable physics-projection layer (Layer B) prior to auditing by a safety model (Layer C). The baseline physics representation—relying on a constant-friction ellipse, a bicycle model, and forward-Euler integration—is vastly insufficient for the highly nonlinear dynamics of limit-handling motorsport. To construct a truly predictive race engineer capable of analyzing adaptive, veteran, and grassroots drivers, the projection layer must map the maximal physics frontier.

This research delineates the absolute ceiling of vehicle dynamics modeling applicable within a 12-day build cycle. The development is governed by absolute hard constraints: the projection layer must remain strictly differentiable to allow continuous gradient flow for the optimization manifold; the implementation must be entirely in Python utilizing automatic differentiation frameworks (PyTorch or JAX); computational execution is limited to consumer-grade hardware (one RTX 4060 laptop, one M2 MacBook); and the development bandwidth is restricted to a two-person engineering team. These constraints necessitate highly vectorized, memory-efficient tensor operations, eschewing bespoke CUDA kernels in favor of natively differentiable operations. What follows is the exhaustive, eight-tier mathematical architecture required to clamp foundation model forecasts onto a rigorous physical manifold.

1. FULL PACEJKA "MAGIC FORMULA" TIRE MODEL

The transition from a theoretical, constant-coefficient friction ellipse to the semi-empirical Pacejka "Magic Formula" (MF) constitutes the most vital structural upgrade to the projection layer. The Magic Formula—specifically iterations 5.2, 6.1, and 6.2—utilizes complex trigonometric functions to accurately map the nonlinear saturation characteristics of pneumatic racing tires across pure and combined slip states.¹

a) Governing Equations

The foundational logic of the Pacejka model asserts that longitudinal force (F_x) and lateral force (F_y) are functions of slip ratio (κ) and slip angle (α), respectively. The pure slip baseline forces are formulated through the core sine and arctangent structure:

$$F_{y0} = D_y \sin (C_y \arctan (B_y \alpha_y - E_y (B_y \alpha_y - \arctan(B_y \alpha_y)))) + S_{Vy}$$

$$F_{x0} = D_x \sin (C_x \arctan (B_x \kappa_x - E_x (B_x \kappa_x - \arctan(B_x \kappa_x)))) + S_{Vx}$$

Where α_y and κ_x represent the effective kinematic slips, which include horizontal shifts (S_H) induced by tire ply-steer and suspension camber.⁴ The structural parameters dictate the exact shape of the force curve: B defines the slip stiffness, C represents the shape factor, D is the peak friction limit, and E is the curvature factor that dictates the sharpness of the peak and the subsequent post-peak falloff.⁶

Crucially, the peak friction D is not static; it exhibits extreme load sensitivity. As the vertical load (F_z) increases, the absolute grip increases, but the coefficient of friction ($\mu = D/F_z$) decreases nonlinearly.⁶ This load dependency is modeled as:

$$D_y = \mu_y F_z = (p_{Dy1} + p_{Dy2} df_z)(1 - p_{Dy3} \gamma^2) F_z$$

Where $df_z = (F_z - F_{z0})/F_{z0}$ is the normalized change in vertical load relative to a nominal rated load F_{z0} , and γ is the inclination or camber angle of the wheel.⁴

Furthermore, racing tires rarely operate in pure slip. Trail-braking into a corner generates combined slip. The theoretical friction ellipse overestimates available grip in these scenarios because the true bivariate friction limit resembles an inverted "heart shape".¹² To enforce this physical reality, the Magic Formula applies differentiable scaling multipliers, G_{xa} and G_{yk} , to the pure slip forces²:

$$G_{yk} = \cos (C_{yk} \arctan (B_{yk} \kappa_x - E_{yk} (B_{yk} \kappa_x - \arctan(B_{yk} \kappa_x))))$$

$$F_y = G_{yk} F_{y0}$$

A parallel cosine-arctangent scaling function G_{xa} is applied to the longitudinal force as a function of the lateral slip angle α , dynamically eroding acceleration potential as the steering wheel is turned.¹⁵

Parameter	Role / Definition	Typical Racing Slick Range	Physical Implication
b_0 / p_{Cy1}	Shape factor	1.4 – 1.8	Defines the asymptotic behavior and falloff after the peak friction. ⁶
b_2 / p_{Dy1}	Friction coefficient peak	1.100 – 1.700	The absolute maximum μ multiplier under nominal load. ⁶
b_4 / p_{Ey1}	Curvature factor	-20.0 to +1.0	Determines the "sharpness" or forgiveness of the tire when breaking traction. ⁶
p_{Dy2}	Load influence on friction	-0.080 to +0.080	Dictates the degressive loss of relative grip as the chassis rolls. ⁶

b) Required Telemetry and Inputs

Implementation demands dynamic vertical load F_z (supplied by the load transfer module), longitudinal velocity v_x , lateral velocity v_y (to compute raw slip angle $\alpha = \arctan(v_y/v_x)$), individual wheel rotational velocities ω (to compute slip ratio $\kappa = (R_w\omega - v_x)/v_x$), and static geometry parameters such as camber γ .¹ The required parameters (the a , b , and p coefficients) are typically sourced from .tir property files or the Formula SAE Tire Test Consortium (FSAE TTC) databases.¹⁸

c) Differentiability Verdict

Fully differentiable. The Pacejka equations consist entirely of smooth transcendental functions ($\sin$, $\cos$, $\arctan$), which are natively supported by automatic differentiation engines in

PyTorch and JAX.²¹ However, the kinematic definitions of slip angle and slip ratio place the longitudinal velocity v_x in the denominator. As velocity approaches zero (e.g., pit lane exits or a spun vehicle), this creates an infinite gradient singularity that will instantly crash a gradient-based optimization solver.³ To guarantee continuous gradient flow and numerical stability within the RTX 4060's VRAM limits, a low-speed regularization constant ϵ must be introduced (e.g., $\alpha = \arctan(v_y/(v_x + \epsilon))$).⁵

d) Fidelity Gained

The integration of a load-sensitive, combined-slip Magic Formula completely eliminates the primary mode of "kinetic hallucination." By clamping the foundation model's output to a heart-shaped combined-slip manifold, the projection layer structurally prohibits the AI from forecasting a lap where a driver simultaneously pulls 1.5G of braking and 1.5G of cornering. It forces the time-series forecast to respect the true tradeoff of trail-braking, yielding highly accurate, physically constrained trajectory limits.

2. TRANSIENT TIRE DYNAMICS

A critical flaw in basic vehicle modeling is the assumption that pneumatic tires act as instantaneous force generators. In reality, the complex viscoelastic structure of the tire carcass and tread band must physically deform against the asphalt before steady-state forces are realized.¹ This delay between a kinematic slip input (e.g., a steering angle change) and the corresponding lateral force response is quantified by the "relaxation length".²⁴

a) Governing Equations

Transient tire dynamics dictate that the forces computed by the Pacejka model cannot be based directly on the instantaneous kinematic slip. Instead, they must be driven by a transient, delayed slip state.⁵ The transient slip angle α' (the effective slip angle genuinely experienced by the contact patch) is modeled via a first-order ordinary differential equation (ODE) based on the "stretched string" theory:

$$\frac{L_y}{v_x} \dot{\alpha}' + \alpha' = \alpha$$

Where L_y is the lateral relaxation length, v_x is the longitudinal velocity, and α is the raw geometric slip angle derived from chassis kinematics.⁵ The lateral force is subsequently calculated by substituting the transient slip angle α' into the Pacejka steady-state functions:

$$F_y = F_{y,steady}(\alpha', F_z, \gamma)$$

The relaxation length itself is not static; it is highly dependent on the operational state of the tire. Empirical testing demonstrates that relaxation length varies with vertical normal load and vehicle speed.²⁷ This dynamic variation is commonly captured via a low-order polynomial mapping:

$$L_y = p_{L1} + p_{L2}F_z + p_{L3}v_x$$

A parallel differential equation governs the transient longitudinal slip ratio κ' using the longitudinal relaxation length L_x , representing the wind-up of the tire sidewall under heavy torque applications.⁵

b) Required Telemetry and Inputs

The transient model requires the instantaneous longitudinal velocity v_x , the raw kinematic slip variables α and κ , and the dynamic vertical load F_z .⁵ Furthermore, the temporal integration of the ODE requires a defined time step Δt , derived from the telemetry logger's frequency.

c) Differentiability Verdict

Fully differentiable, provided the numerical integration scheme is carefully chosen. ODEs can be naturally embedded into PyTorch or JAX computation graphs.²¹ However, the time constant of the tire ODE is $\tau = L_y/v_x$. At low speeds, this time constant becomes exceedingly large, transforming the system into a stiff differential equation.³ Applying a standard, explicit Forward-Euler integration (as used in the V1 model) to a stiff ODE will lead to severe numerical oscillations and exploding gradients during backpropagation. The implementation must utilize a differentiable implicit solver (such as Backward Euler) or leverage neural ODE toolkits (e.g., torchdiffeq in PyTorch) to compute the forward pass while preserving continuous, stable gradient flow.²³

d) Fidelity Gained

The inclusion of transient dynamics corrects phase-shift errors during high-frequency steering inputs, such as navigating a slalom or a rapid chicane.²⁶ Without relaxation lengths, the AI would forecast mathematically permissible but physically impossible immediate spikes in lateral acceleration. Transient modeling enforces a continuous low-pass filter on the vehicle's dynamic response, ensuring the forecasting envelope perfectly respects the true physical inertia,

carcass stiffness, and yaw damping of the real racing car.

3. TIRE THERMAL & DEGRADATION MODELING

The mechanical friction limits defined by the Pacejka equations are highly coupled to the thermodynamic state of the rubber compound. Cold racing slicks exhibit brittle, low-friction characteristics, while excessively hot tires suffer from thermal degradation, blistering, and an abrupt drop in grip.³² A robust projection layer must track the thermal energy balance of the tire, translating slip kinematics into heat generation, and modulating the absolute friction envelope accordingly.³³

a) Governing Equations

A computationally efficient yet highly accurate thermal model divides the tire into two interacting lumped thermal masses: the outer tread surface (T_t) and the internal carcass (T_c).³⁵ The First Law of Thermodynamics provides the governing differential equation for the tread temperature:

$$m_t c_t \dot{T}_t = Q_{gen} - Q_{conv} - Q_{cond} + Q_{carcass} - Q_{rad}$$

Where m_t is the tread mass and c_t is the specific heat capacity of the rubber compound.³⁵ The individual energy fluxes are defined as follows:

1. **Frictional Heat Generation (Q_{gen})**: The mechanical power dissipated at the contact patch generates heat.

$$Q_{gen} = p_1 \cdot (|F_x v_{sx}| + |F_y v_{sy}|)$$

Here, v_{sx} and v_{sy} are the longitudinal and lateral slip velocities, and p_1 is a tunable partition coefficient (typically ~0.5) that represents the specific fraction of total frictional power absorbed directly by the tread, rather than dissipated as acoustic energy or track wear.³⁵

2. **Convective Cooling (Q_{conv})**: Heat stripped away by the surrounding airflow.

$$Q_{conv} = h_{conv} A_{conv} (T_t - T_{amb})$$

Where the convection coefficient h_{conv} scales dynamically with the relative vehicle velocity, and A_{conv} is the exposed tread area.³⁵

3. **Conductive Cooling** (Q_{cond}): Heat transferred directly into the track surface at the contact patch.

$$Q_{cond} = h_{cond} A_{contact} (T_t - T_{track})$$

Modeled via Newton's law of cooling utilizing an appropriate heat exchange coefficient h_{cond} .³⁸

4. **Internal Conduction** ($Q_{carcass}$): Heat exchanged between the tread and the bulk carcass.

$$Q_{carcass} = h_{internal} A_{total} (T_c - T_t)$$

5. **Radiative Cooling** (Q_{rad}): Heat radiated to the environment, governed by the Stefan-Boltzmann law.

$$Q_{rad} = \epsilon \sigma A_{rad} (T_t^4 - T_{amb}^4)$$

Where ϵ is the emissivity of rubber and σ is the Stefan-Boltzmann constant.³⁵ The ultimate purpose of tracking these temperatures is to dynamically alter the tire's grip. The real-time grip potential is modulated by scaling the Pacejka peak friction coefficient D (specifically the p_{Dy1} and p_{Dx1} terms) according to a thermal distribution curve, often modeled as a parabolic or Gaussian function²⁸:

$$\mu(T_t) = \mu_{peak} - p_{T1} (T_t - T_{opt})^2$$

Where T_{opt} is the optimum operating temperature of the racing slick, and p_{T1} dictates the sharpness of the performance falloff as the tire deviates from this optimum.²⁸

b) Required Telemetry and Inputs

The thermodynamic model requires ambient temperature T_{amb} , track temperature T_{track} , continuous longitudinal velocity v_x , calculated slip velocities v_{sx}, v_{sy} , and the instantaneous

tire forces F_x, F_y .³⁵ Continuous state tracking over the course of a race stint requires the model to preserve hidden states between laps.

c) Differentiability Verdict

Fully differentiable. The thermal differential equations contain only standard algebraic operations and absolute values.³⁵ While the absolute value function $|x|$ has a non-differentiable point at exactly zero, modern frameworks like PyTorch and JAX automatically assign sub-gradients (typically 0) at this point, ensuring uninterrupted optimization.²² Furthermore, because the thermal masses of the tread and carcass possess relatively large time constants, temperature evolves slowly compared to kinematic motions. Consequently, simple integration schemes are highly numerically stable and permit unhindered backpropagation without exploding gradients.

d) Fidelity Gained

The inclusion of thermodynamics unlocks long-horizon forecasting and degradation awareness. Without a thermal state, the AI assumes the tires on lap 15 possess the exact same grip envelope as lap 2. By explicitly coupling mechanical slip power to temperature generation, the projection layer can detect when a driver is overdriving or sliding excessively, correctly penalizing the available friction budget and forecasting inevitably slower lap times as the tires overheat.³²

4. LOAD TRANSFER

The ability of any individual tire to generate force is intrinsically dependent on its instantaneous vertical normal load, F_z . As a vehicle accelerates, brakes, or corners, the inertial forces acting through the chassis's center of gravity (CG) create overturning moments. These moments transfer load from the inner to the outer tires, and from the front to the rear tires.⁴¹ Because tire friction does not scale linearly with load (due to the Pacejka load sensitivity parameters outlined in Section 1), accurately mapping this load transfer is paramount.

a) Governing Equations

Total load transfer must be mathematically separated into its longitudinal and lateral vectors.

For a rigid-body chassis, the total longitudinal load transfer $\Delta F_{z,lon}$ under acceleration a_x is a function of the vehicle's geometry:

$$\Delta F_{z,lon} = \frac{m \cdot a_x \cdot h_{cg}}{L}$$

Where m is the total vehicle mass, h_{cg} is the height of the center of gravity, and L is the

wheelbase.¹⁰

Lateral load transfer is vastly more complex, as it relies on the specific kinematics of the suspension geometry and the distribution of elastic stiffness.⁴² The total lateral load transfer across a given axle is the sum of three distinct physical components: the unsprung mass transfer, the sprung mass geometric (kinematic) transfer, and the sprung mass elastic transfer.¹⁰

The elastic component—which is highly tunable via suspension springs and anti-roll bars (ARBs)—is dictated by the vehicle's roll stiffness distribution. For the front axle, the elastic load transfer is governed by:

$$\Delta F_{z,lat,elastic,front} = \left(\frac{K_{\phi,f}}{K_{\phi,f} + K_{\phi,r}} \right) \frac{m_{sprung} \cdot a_y \cdot h_{roll}}{t_f}$$

Where $K_{\phi,f}$ and $K_{\phi,r}$ represent the front and rear suspension roll stiffnesses (Nm/deg), a_y is the lateral acceleration, h_{roll} is the vertical distance between the CG and the chassis roll axis, and t_f is the front track width.⁴²

Concurrently, the geometric load transfer passes directly through the physical suspension arms, bypassing the springs entirely. This is dictated by the roll center heights ($h_{rc,f}$ and $h_{rc,r}$) determined by the suspension hardpoints:

$$\Delta F_{z,lat,geom,front} = \frac{m_{sprung} \cdot a_y \cdot h_{rc,f}}{t_f} \cdot \left(\frac{l_r}{L} \right)$$

Where l_r is the longitudinal distance from the CG to the rear axle.⁴⁴

b) Required Telemetry and Inputs

The load transfer module requires continuous longitudinal and lateral accelerations a_x, a_y sourced directly from the IMU telemetry. Additionally, it requires an array of static vehicle parameters: total mass m , unsprung mass fractions, wheelbase L , front and rear track widths t_f, t_r , CG height h_{cg} , roll center heights $h_{rc,f}, h_{rc,r}$, and the specific roll stiffnesses $K_{\phi,f}, K_{\phi,r}$.⁴²

c) Differentiability Verdict

Fully differentiable. The load transfer models consist purely of linear algebraic expressions and fraction distributions.²¹ Within PyTorch or JAX, these operations are trivial to differentiate. More importantly, these purely linear operations can be executed across massive time-series tensors simultaneously via batch vectorization operations (vmap), generating dynamic F_z states for all four wheels across an entire lap with negligible computational overhead on the host GPU.²²

d) Fidelity Gained

The inclusion of detailed sprung and unsprung load transfer effectively bridges the theoretical gap between suspension tuning and absolute tire grip. Because tire grip scales degressively with load, transferring weight from the inner to the outer tire fundamentally reduces the total aggregate grip of the axle.⁹ This layer is what allows the AI to recommend highly specific physical tuning changes—such as stiffening the rear anti-roll bar—to dynamically alter the load transfer distribution and cure aerodynamic or mechanical understeer.

5. BEYOND THE BICYCLE MODEL

The deliberately minimal V1 system utilizes a "bicycle model," which mathematically collapses the left and right wheels of the vehicle into a single central track.⁴⁴ While computationally convenient for analyzing basic yaw dynamics and steady-state steering responses, a single-track model completely ignores the asymmetrical load distribution during cornering, as well as the distinct, opposing slip angles generated by the inner and outer wheels.⁴² Upgrading the projection layer to a four-wheel, double-track model is mandatory to capture true limit-handling behavior.

a) Governing Equations

The double-track model fundamentally calculates individual normal loads F_z for all four contact patches by superimposing the static static weight distribution with the dynamic load transfer components calculated in Section 4⁴²:

$$F_{z,front-left} = \frac{m \cdot g \cdot l_r}{2L} - \frac{\Delta F_{z,lon}}{2} + \Delta F_{z,lat,front}$$

$$F_{z,front-right} = \frac{m \cdot g \cdot l_r}{2L} - \frac{\Delta F_{z,lon}}{2} - \Delta F_{z,lat,front}$$

$$F_{z,rear-left} = \frac{m \cdot g \cdot l_f}{2L} + \frac{\Delta F_{z,lon}}{2} + \Delta F_{z,lat,rear}$$

$$F_{z, rear-right} = \frac{m \cdot g \cdot l_f}{2L} + \frac{\Delta F_{z, lon}}{2} - \Delta F_{z, lat, rear}$$

In addition to highly separated normal loads, the double-track model incorporates Ackermann steering geometry. During a corner, the inner front wheel traces a tighter geometric radius than the outer front wheel, necessitating a larger steering angle to prevent tire scrubbing.⁴⁷ The

distinct steering angles δ_{in} and δ_{out} determine the independent kinematic slip angles for the front wheels:

$$\alpha_{front-left} = \delta_{front-left} - \arctan \left(\frac{v_y + l_f \dot{\psi}}{v_x - \frac{t_f}{2} \dot{\psi}} \right)$$

$$\alpha_{front-right} = \delta_{front-right} - \arctan \left(\frac{v_y + l_f \dot{\psi}}{v_x + \frac{t_f}{2} \dot{\psi}} \right)$$

Where $\dot{\psi}$ is the chassis yaw rate.⁴⁴ The total lateral force and the restorative yaw moment acting on the chassis are then resolved by explicitly summing the four individual, independent Pacejka tire forces:

$$\sum F_y = F_{y, fl} \cos(\delta_{fl}) + F_{y, fr} \cos(\delta_{fr}) + F_{y, rl} + F_{y, rr}$$

b) Required Telemetry and Inputs

This expansion requires the chassis yaw rate $\dot{\psi}$ (from the gyroscope IMU), the driver's steering wheel angle, the steering rack ratio mapping (to compute individual wheel angles), and precise vehicle dimensions (track widths t_f, t_r , and the longitudinal distance from the CG to the axles l_f, l_r).⁴⁴

c) Differentiability Verdict

Fully differentiable. The transition from a two-wheel to a four-wheel model introduces no novel mathematical complexities; it merely expands the dimensionality of the tensor operations from shape $[N, 2]$ to $[N, 4]$. The trigonometric additions required for Ackermann geometry and slip

angle differentiation are entirely natively supported by the PyTorch and JAX autograd engines.²²

d) Fidelity Gained

The double-track model provides the mechanical mechanism for accurately forecasting severe oversteer and understeer limits. By resolving the four tires independently, the projection layer can accurately model limit-handling scenarios where the inside tire unloads entirely or lifts

completely off the ground ($F_z = 0$), forcing the outside tire to absorb the entirety of the lateral load. This prevents the foundation model from hallucinating higher cornering capabilities than a severely rolling chassis could actually support, leading to superior safety audits.

6. 3D TRACK GEOMETRY

Standard vehicle dynamics algorithms, including the V1 system, inherently assume the car is driving on an infinite, flat, 2D plane. In both grassroots and professional motorsport, real-world track topography—such as the massively banked carousel at the Nürburgring, the violent elevation change of Eau Rouge at Spa, or simply heavily crowned road surfaces—drastically alters the gravity vector relative to the vehicle, subsequently modifying the available friction limit.⁴⁹

a) Governing Equations

To mathematically correct the 2D friction ellipse for 3D surfaces, the system must translate world-frame gravity into the vehicle coordinate system utilizing 3D Euler angle rotations (Yaw

ψ , Pitch θ , Roll ϕ).⁵² The rotation matrix transforms the standard gravity vector $g = [0, 0, -9.81]^T$ directly into the vehicle's reference frame:

$$R = R_z(\psi)R_y(\theta)R_x(\phi)$$

The effective normal force contribution from gravity is dynamically altered by the road banking (roll) and gradient (pitch)⁵³:

$$g'_z = g \cos(\theta) \cos(\phi)$$

$$g'_y = g \cos(\theta) \sin(\phi)$$

$$g'_x = g \sin(\theta)$$

In a highly banked corner, the lateral component of gravity physically assists the tires in reacting to the centrifugal force. The effective lateral force balance equation becomes ⁵¹:

$$F_{y,tires} + m \cdot g \cos(\theta) \sin(\phi) = m \cdot a_{y,kinematic}$$

Furthermore, vertical track curvature (e.g., dipping into a compression zone) creates a massive additional centripetal acceleration vector acting vertically on the car, artificially augmenting the normal load:

$$F_{z,static} = m \left(g'_z + \frac{v_x^2}{R_{vertical}} \right)$$

b) Required Telemetry and Inputs

Implementation requires high-frequency (100 Hz) IMU data (3-axis accelerometers, 3-axis gyroscopes) and GPS telemetry (latitude, longitude, altitude).⁵⁵ Because budget data loggers suffer from extreme noise, the 3D track geometry is reconstructed offline prior to the forecast by passing the GPS/IMU telemetry through a Kalman filter or a complementary filter to estimate the continuous road pitch and bank angles as a function of track distance.⁵⁶

c) Differentiability Verdict

Fully differentiable. The reconstruction of the track geometry is a pre-processing state-estimation step executed once per session. Inside the optimization loop of the projection

layer, the track angles (ϕ, θ) are injected as fixed tensors aligned to the track distance index. The coordinate rotations and gravity distribution functions are simple, smooth trigonometric matrix multiplications handled with extreme efficiency by PyTorch or JAX.²¹

d) Fidelity Gained

This layer corrects "kinetic hallucinations" driven by extreme topographies. A car pulling 2.0G of lateral acceleration on a perfectly flat surface might violently violate Pacejka physics, prompting the Guardian layer to flag a catastrophic physics failure. However, if the track is banked at 18 degrees, the banking absorbs a significant portion of the lateral load, keeping the tires well within their friction circle. Injecting 3D geometry entirely eliminates these false-positive safety violations.

7. AERODYNAMICS

While grassroots racing features fewer downforce-producing elements than Formula 1, vehicles utilizing front splitters, rear wings, and underbody diffusers exhibit intense speed-dependent load properties. Aerodynamic downforce artificially increases the vertical normal load F_z on the tires without adding corresponding inertial mass, drastically expanding the Pacejka friction envelope at high speeds.⁵⁹

a) Governing Equations

Total aerodynamic downforce $F_{z,aero}$ and aerodynamic drag $F_{x,aero}$ are proportional to the square of the vehicle's longitudinal velocity⁵⁹:

$$F_{z,aero} = \frac{1}{2} \rho \cdot v_x^2 \cdot A \cdot C_L$$

$$F_{x,aero} = \frac{1}{2} \rho \cdot v_x^2 \cdot A \cdot C_D$$

Where ρ is air density, A is the frontal reference area, and C_L, C_D are the aerodynamic lift (downforce) and drag coefficients.⁶⁰ The downforce must be physically distributed to the front and rear axles based on the aerodynamic balance (the Center of Pressure, or CoP).⁶² Crucially, aerodynamic downforce is highly sensitive to the pitch angle of the car.⁵⁹ Under heavy braking, the car dives (negative pitch), bringing the front splitter closer to the ground and significantly increasing the front lift coefficient $C_{L,f}$, while simultaneously stalling the rear wing. This critical pitch sensitivity can be modeled as a linear approximation:

$$C_{L,total}(\theta) = C_{L0} + \frac{\partial C_L}{\partial \theta} \cdot \theta_{pitch}$$

$$\text{Aero Balance}(\theta) = \text{Bal}_0 + \frac{\partial \text{Bal}}{\partial \theta} \cdot \theta_{pitch}$$

The computed aerodynamic downforce is super-positioned onto the dynamic vertical load calculation (from Section 4) prior to passing the final F_z tensor into the Pacejka model.⁴⁴

Parameter	Symbol	Value Example	Physical Implication
Downforce Coefficient	C_{L0}	-2.5 to -4.8	The baseline chassis downforce generated at ride height. ⁵⁹
Drag Coefficient	C_D	0.8 to 1.5	The parasitic aerodynamic drag counteracting longitudinal acceleration. ⁶²
Aero Balance	Bal_0	40% to 58%	The percentage of total downforce acting on the front axle. ⁵⁹
Pitch Sensitivity	$\frac{\partial C_L}{\partial \theta}$	0.12 per degree	The dynamic loss/gain of downforce as the suspension pitches. ⁵⁹

b) Required Telemetry and Inputs

Requires velocity v_x , and an estimated pitch angle θ_{pitch} derived mathematically from the longitudinal load transfer and the suspension spring stiffness. Ambient weather data

(temperature, barometric pressure) is necessary to determine air density ρ . Static wind-tunnel or CFD-derived parameters (C_L, C_D, CoP) are configured per-vehicle.⁵⁹

c) Differentiability Verdict

Fully differentiable. Aerodynamics introduces a quadratic dependency on velocity and a linear dependency on pitch angle. These are standard polynomial operations easily and efficiently traced by Autograd in PyTorch or JAX²², demanding virtually no extra memory overhead.

d) Fidelity Gained

Aerodynamics accurately shifts the friction boundary dynamically with speed, ensuring that a high-speed corner forecast correctly permits significantly higher lateral Gs than a low-speed

hairpin. The pitch-sensitivity mapping is paramount; it allows the projection layer to anticipate mid-corner instability (entry oversteer) caused by the aero balance shifting dangerously forward under trail braking.

8. ADAPTIVE / HAND-CONTROL VEHICLE DYNAMICS

Disabled and veteran drivers utilizing FIA-certified electronic hand controls employ vastly unique driving techniques compared to able-bodied drivers. Standard vehicle physics models

operate on the assumption that throttle and brake inputs are mutually exclusive ($T \times B = 0$). Hand controls, however, allow for independent, simultaneous actuation of the throttle and brake.⁶⁵ This simultaneous input radically alters longitudinal dynamics, differential locking states, and suspension pitch behavior.

a) Governing Equations

When brake and throttle are applied concurrently, the total longitudinal force demanded at the contact patch of the driven wheels is a superposition of the positive driving torque T_{eng} and the negative braking torque T_{brake} .⁶⁶ For a standard rear-wheel-drive vehicle:

$$F_{x,rear} = \frac{T_{eng} \cdot R_{gear} \cdot R_{final}}{R_w} - \frac{T_{brake,rear}}{R_w}$$

Where R_w is the tire radius. This simultaneous input creates three critical dynamic conflicts that must be mathematically resolved:

1. **Friction Budget Consumption:** The Pacejka combined slip multiplier (Section 1) is

dictated by the *net* longitudinal slip κ . However, the thermodynamic heat generation (Section 3) is driven by the gross opposing mechanical energies. The brakes and engine are fighting each other, consuming the thermal budget rapidly.

2. **Differential Locking:** A mechanical limited-slip differential (LSD) locks its clutch plates based on input torque. Simultaneous braking and throttling apply massive opposing torques to the axles, artificially locking the differential. The yaw moment bias from the differential $\Delta M_{z,diff}$ becomes a function of both brake pressure and throttle position, violently inducing understeer:

$$\Delta M_{z,diff} = \text{sgn}(\Delta\omega) \cdot T_{lock} (|T_{eng}| + |T_{brake}|)$$

3. **Pitch Conflict:** Engine torque pushes the chassis forward via the rear suspension anti-squat geometry, while rear brakes pull the chassis backward via anti-lift geometry. This simultaneous activation binds the suspension linkages, effectively multiplying the

rear roll stiffness $K_{\phi,r}$ and altering the lateral load transfer distribution calculated in Section 4.

b) Required Telemetry and Inputs

Requires throttle position percentage, brake pressure (bar), engine RPM, gear state, steering angle, and an explicit flag detailing the differential type (e.g., 1.5-way LSD, open, or electronic torque-vectoring).⁶⁵

c) Differentiability Verdict

Fully differentiable, but requires specific algorithmic smoothing. The superposition of forces is a basic linear addition. However, to model the differential locking mechanism, the

non-differentiable absolute value function ($|T|$) and the discontinuous sign function (sgn) must be approximated using continuously differentiable equivalents (e.g.,

$\text{sgn}(x) \approx \tanh(k \cdot x)$ where k is a large steepness scalar).²² This guarantees uninterrupted gradient flow during sudden differential locking state changes.

d) Fidelity Gained

This is the ultimate, foundational differentiator for the "PIT WALL" architecture. While standard dynamic planners reject simultaneous inputs as anomalies or telemetry errors, this layer quantifies exactly how dragging the brake against the throttle stabilizes the vehicle's platform by artificially locking the differential and pre-loading the suspension. It allows the system to offer structural tuning recommendations specifically tailored to adaptive driving techniques, rather than trying to force adaptive drivers into restrictive, able-bodied physics models.

MAXIMAL PHYSICS STACK

To ensure numerical stability, thermodynamic continuity, and proper forward-pass logic within the differentiable tensor framework, the eight domains must be layered in the following strict execution sequence inside the optimization projection layer:

Layer 1: 3D Track Geometry

Ingests spatial location and IMU telemetry to project the global gravity vector and establish the true road curvature, setting the baseline for all subsequent vertical load calculations.

Layer 2: Aerodynamics

Calculates speed-dependent downforce and drag, and dynamically distributes the aero load across the front and rear axles based on instantaneous pitch approximations.

Layer 3: Adaptive / Hand-Control Dynamics

Resolves simultaneous throttle/brake inputs into net longitudinal forces and calculates the mechanical binding state of the suspension and the limited-slip differential.

Layer 4: Load Transfer (Beyond the Bicycle Model)

Executes the double-track geometry to distribute the combined static, aerodynamic, and inertial loads into four distinct normal forces (F_z), factoring in roll stiffness distribution.

Layer 5: Tire Thermal & Degradation

Tracks the generation and dissipation of heat based on the previous frame's slip velocities,

updating the real-time peak friction potential (μ_{peak}) for the current forecasting frame.

Layer 6: Transient Tire Dynamics

Applies relaxation length delays to the raw kinematic slip angles, filtering out instantaneous changes and extracting the true transient slip states experienced by the contact patches.

Layer 7: Full Pacejka Tire Model

Receives the dynamic F_z , thermal μ_{peak} , and transient slip states to compute the final, load-sensitive, combined-slip lateral and longitudinal forces based on the inverted heart-shape boundary.

Layer 8: Vehicle Kinematic Integration

Sums the forces from Layer 7, applies Newton's second law, and resolves the final, physically constrained vehicle accelerations to be audited by the IBM Granite Guardian model.

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